CONTRACEPTION

Definition: Prevention of pregnancy at will.

Methods:

1- Physiological methods.

2- Barrier methods.

3-Chemical methods.

4- Intrauterine contraceptive device (I.U.C.D).

5- Hormonal methods.

6- Sterilization.

I - Physiological Methods

[Natural family planning method]

[1] Lactation:

Failure rate: 5-40/HWY.(hundred women/years) i.e. percentage of pregnancy which occurs with 100 females using this methods of contraception for 1 year.

Mechanism: -With breast feeding → increase prolactin → decrease gonadotropin hormone → amenorrhea. (Increase PRL α decrease FSH).

- But to ↑ efficiency of this method: (6-8) feedings / day, each time 10 min. at least, with midnight feeding.

[2] Safe period: = (Fertility awareness method)

Failure rate = 2-20 / HWY.

Mechanism:- Based on avoiding coitus during fertile time in menstrual cycle.

- We can measure it by:

A) *Calendar method*:

- Record of 6-12 cycles is done.
- The shortest and longest cycles are defined.
- 1st day of the potentially fertile period is the shortest cycle minus 20 days.

- Last day of the potentially fertile period is the longest cycle minus 10 days e.g. regular cycle 28 days.

  1st day = 28-20 = 8

  Last day = 28 – 10 = 18

  So the fertility period (unsafe period) = 8-18. So the safe period, before day 8, or after day 18.

B) *B.B.T. Method:*

- It relies on the thermogenic effect of progesterone in the second half of cycle.
- Female is considered fertile from the beginning of the cycle till 3 consecutive days after the rise in the temperature.

C) *Cervical mucus method:*

- Around the time of ovulation, the cervical mucus is abundant, clear and stretchable. So coitus is permitted in dry days of the cycle.
- The coitus is permitted in days following menses (bleeding) till female is wet around ovulation, then coitus is resumed when dryness return again (7days later)

D) *Symptothermal method:*

Combination of

Symptoms + Thermal method

↓ ↓

Cx. Mucus BBT-Chart

E)*Double – check method:* use 2 methods together.

[3] Coitus Interruptus:

Failure rate = 10-30/ HWY.

Mechanism: Based on interruption of coitus and withdrawal of male organ just before orgasm with external ejaculation of semen away from genitalia.

Disadvantages:

1- Unreliable with ↑ failure rate due to:

a- Pre-ejaculatory semen is very rich in sperms. b- Lack of self control.

2- Marital , sexual and psychological upset.

3- Chronic pelvic congestion.

Advantages:

1- No cost.

2- Always available.

3- No need for chemicals.

II- Barrier Methods

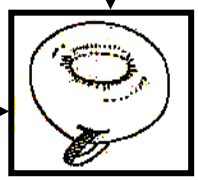
[mechanical methods]

A- Male

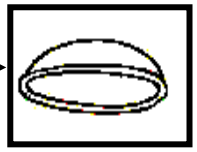
Condom

B- Female

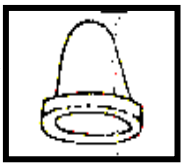
-V. sponge



-V. diaphragm



- Cervical cap



- Female condom

Male condom (French letter):

* **Types:**

1- Synthetic polyethylene fine latex (commonly used) 3-3.5 cm diameter, 15-20 cm long & 0.03-0.07 thick

- One closed teat-shaped end.
- Disposable, sterile, coloured with lubrication by spermicidal.
- Used on erect penis before penetration.

2- Washable sheaths made of thick rubber (not used).

3- Anti Allergic type for ♂ with hypersensitivity.

4- Animal sheath from caecum, expensive.

5- Glans- covering condoms (American tips) with ↑ failure rate.

6- Urethral insertion of smell bags.

* **Effectiveness:** - Method failure 1-5% - Use failure 5-15%

* **Indications:**

1- Postpartum till other method is used.

2- As an additional method for contraception.

3- When other methods are unaccepted.

***Contraindications:**

1- Anatomically malformed penis.

2- Allergy to rubber in ♂ or ♀.

3- Psychologically unaccepted.

* **Advantages:**

1- ***Contraceptive:*** effective when used properly, cheap, few side effects, need no medical supervision.

2- ***Non contraceptive advantages:*** protect against STD, PID, ↓ risk for CX cancer, ↓ premature ejaculation , help to maintain erection, used in treatment of immunological infertility.

* **Disadvantages:**

1- Interrupt love- making.

2- ↓↓ pleasurable sensation.

3- Allergy.

4- Increase failure rate.

* **Instructions:**

1- Adjuvant spermicidal.

2- Used on erect penis

3- Held firmly during withdrawal.

4- No Vaseline (as ↓ tensile strength).

5- Never to use disposable again.

6- If Torn → use post coital contraception.

Diaphragm (Dutch cap)

* **Types:**

1- ***Flat spring:*** Firm → remains horizontal on compression, it's easily fitted, suitable for normal vagina.

2- ***Coil spring:*** Softer → difficult to control on insertion, suitable for pt. sensitive to pressure.

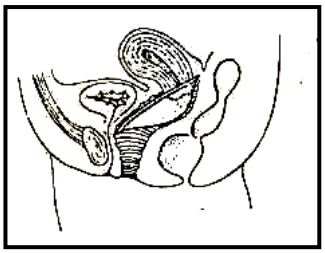
3- ***Arcing diaphragm:*** combines features of flat & coil spring.

4- Recent disposable spermicidal coated diaphragm:

* **Effective ness:** failure rate 5-15/ Hwy.

* **Indications:** as condom.

* **Contraindications**:



1- Anatomically malformed ♀ G.T.

2- Lack of hygiene with recurrent U.T.I.

3- Latex rubber allergy.

4- Psychological factor.

* **Advantages:**

1- As contraception: Cheap, simple, no medical supervision with little side effects.

2- As non-contraceptive advantage: when used e^- spermicidal it can ↓ PID & STD.

* **Disadvantages:**

1- As condom +

2- T.S.S if retained for a time

3- ↑ dysuria & U.T.I.

* **Instructions:**

1- Used e^- every coitus (before coitus) with spermicidal.

2- Repeated coitus → needs repeated spermicide.

3- Must be left for > 6^{hr} after coitus but not > 24^{hr}.

4- After used cleaned, dried, stored in cool dry area.

5- Check for a hole before every use.

Vaginal Contraceptive Sponge (Today- Sponge)

* It's ployurethrane soft white sponge, 5.5 cm in diameter with loop for pulling from it, contain 1gm nonoxynol-9.
* Effectiveness: 5-15 / HWY failure rate
* Side effects: - Sensitivity to spermicidal.
  - Irritation & ↑ fungal infection.
  - T.S.S. if retained > 2 days.
  - High failure rate.

Cervical cap.

* **Types:**

1- Cavity rim cap: Size: diameter 22, 25 mm , 28, 31, on normal Cx.

2- Vault cap. (Dumas): size: 55-75 mm on short Cx.

3- Vimule cap : Size : 3 sizes 45, 48, 51 on long Cx.

4- Contraceptive recent cap: has one way valve → allow passage of Cx mucus & menstruation not the sperms to be left in place for 3 months.

* **Side effects:**

1- More difficult in position & wear.

2- Can cause Cx infection & ulcers.

3- Accidental dislodgement → failure.

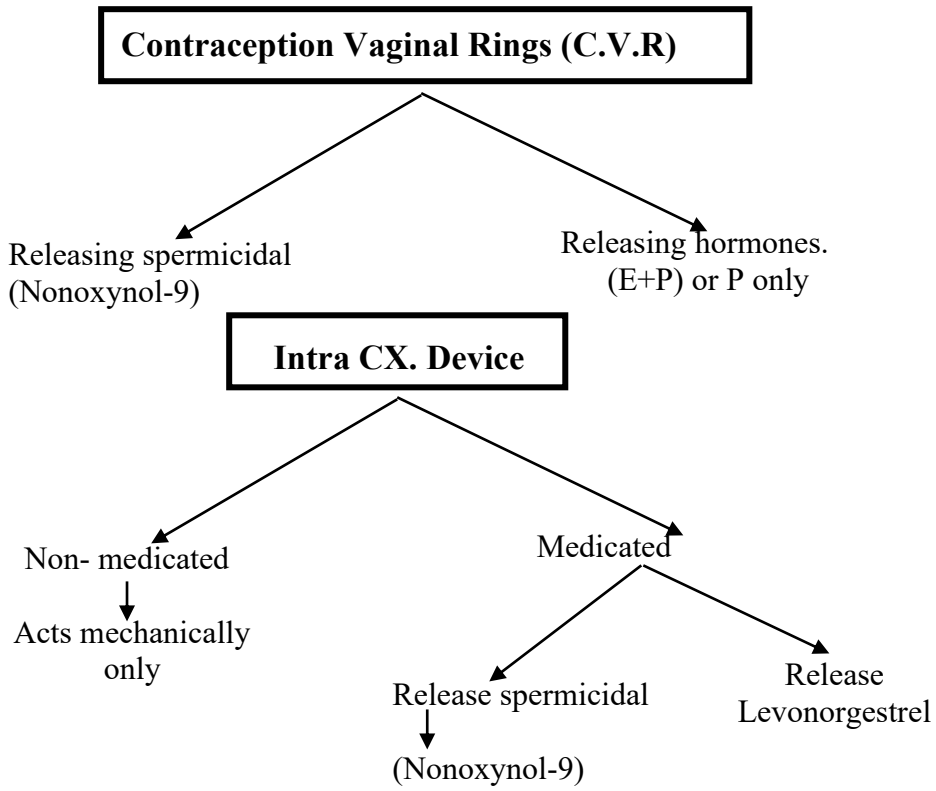
Female Condom (Femshield)

* Soft polyurethane sheath, 15 cm long, covers the vagina & vulva, has 2 rings: outer 7 cm covers ext. genitalia, inner 6.5 Cm.
* Advantages:

1. Stronger, less liable to break.
2. More accepted.
3. No slipping in the vagina.
4. ↓ All STD.
5. Needs no medical supervision.

* Can be used for selected gynecological indications:

1- Superficial dyspareunia.
2- During menstruation & puerperuim.
3- Some cases of immunological infertility.



III- Spermicidals

* **Definition:** Chemicals act by inactivation of sperms in vagina before move to upper G.T.

Formed of 2 basic components

Active spermicidal agents

Usually surface acting → destroy sperm cell membrane & ↓ O_2 uptake

e.g. Nonoxynol 9,10,11 and octoxynol-9

e.g. Contra seed (nonoxynol-9 100mg)

Basic (carrier) agent

e.g. creams, foams, foaming tab., suppositories foams & creams better than suppositories being more rapidly distributed.

* **Effectiveness:**

It's not effective when used alone and failure rate up to 30/HWY.

* **Indications:**

1- As adjuvant to other methods.

2- As emergency when condom ruptured.

3- With IUCD to ↑ effectiveness to 100%.

* **Contraindications:**

1- Allergy.

2- Oil- based products, not with condom as it ↓ its tensile strength.

* **Advantages:**

1- No systemic side effects.

2- Gives lubrication e^- coitus.

3- Spermicidal inactivate HIV & other STD.

* **Disadvantages:**

1- ↑ failure rate.

2- Unpleasant odour & some irritation

* **Instructions:**

1- Creams 3 min.
- Foaming tab. 10 min.
- Supp. 15 min

} Before coitus.

2- High-up in vagina

3- Repeated, of coitus repeated.

4- No douching, for 6^{hr} after coitus.

* **New spermicidal:**

1- Chlorhexidine : acts in Cx mucus & vagina.

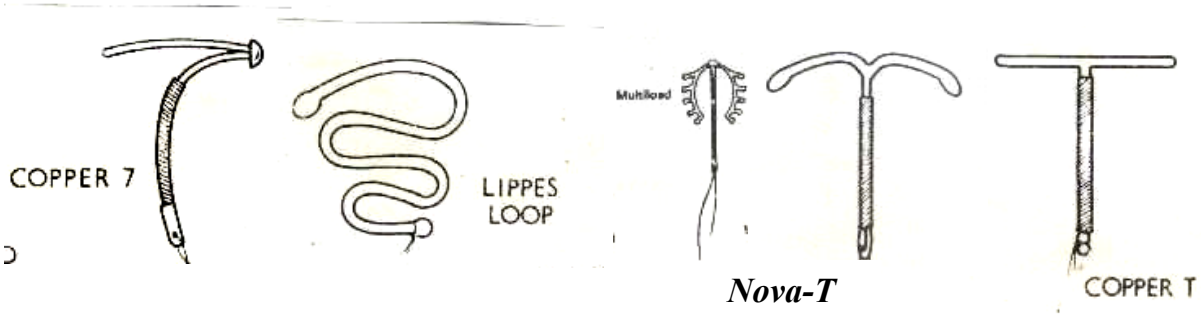
2- Gossypol: acts locally by block acrosomal enzyme.

3- RS- 37367: immobilization of sperms without destruction of cell membrane found to be 50 x potent as Nonoxynol-9.

4- Propronolol (β-blocker):↓ sperm motility & metabolism.

5-Dextrin sulphate: spermicidal + antiviral & anti fungal.

IV- Intrauterine Contraceptive Device (IUCD)



Types:

1-**Non-medicated devices:** These are plastic devices that contain barium sulfate to make them radiopaque e.g. lippes loop (double S-loop) not used now.

2- **Bioactive (Medicated) devices:**

- Copper (Cu) medicated
- Progesterone medicated
- Antibleeding medicated

A) Copper – Medicated:

These contain copper and recently silver. The copper increases contraceptive efficacy of IUCD while silver to decrease rate of (Cu) fragmentation e.g.:

- Cu-**7.**
- Cu T-200, Cu T-380.
- Nova-T (containing silver).
- Multiload (250 - 375).

B) Progesterone-releasing devices: which decrease amount of bleeding due to progesterone.

1- The Mirena LNG-IUS consists of a plastic T-shaped frame with a steroid reservoir around a vertical stem of polydimethylsiloxane. The stem contains 52 mg of levonorgestrel, the laevo – isomer of

norgestrel, derived from the 19-nortestosterone progestogens, released at a rate of 20 μg per day.

2- The progestasert intrauterine progesterone system (PIPS) consists of a polymeric T-shaped platform with a reservoir containing 38 mg of progesterone released at a rate of 65μg per day.

3- The fibro plant LNG IUS consists of a fibrous delivery system of 3 cm length and 1.2 mm diameter that releases 14 μg of levonorgestrel daily. This is fixed to an anchoring filament by means of a metal clip positioned 1 cm from the anchoring knot.

C) Antibleeding medicated: to decrease amount of bleeding

Mode of action of IUCD: The exact mechanism is unknown.

1- It acts as mechanical barrier preventing ovum implantation in the endometrium.
2- It can initiate sterile inflammatory reaction.
3- ↑ endometrial PG.
4- ↑ uterine and tubal motility.
5- Copper interferes with the endometrial enzymes so inhibits early development and implantation of embryo.

Failure rate:
- Lippes loop: 4 / HWY.
- Cu 7, Cu T 200: 2- 4 / HWY.
- In recent types (Cu 380, Multiload) is less than 1/HWY.

Indications of IUCD:

A- <u>Contraceptive:</u>

1- Parous women who need a highly effective reversible method.
2- One decision method for long time.
3- Cheap, prefer in national programs.
4- Breast feeding women.
5- In old age.

B- <u>Non- contraceptive:</u>

1- ttt of Asherman's syndrome to prevent readhesion after cutting it.
2- Progesterone releasing devices for uterine hyperestrogenic condition

Contraindications of IUCD:

A- Absolute contra-indications:

1- Pelvic inflammatory disease (PID).

2- Known or suspected pregnancy.

3- Malignant disease: so Pap. smear for cervical cancer is done routinely before use of IUCD.

B- Relative contra-indications:

1- Nulliparity: due to → Difficult insertion
→ Danger of PID

2- Previous ectopic pregnancy.

3- Abnormal uterine bleeding.

4- Severe anemia.

5- Impaired coagulation response.

6- Vascular heart disease: for fear of subacute bacterial endocarditis.

7- Severe dysmenorrhea.

8- Distorted uterine cavity: e.g. by myoma or bicornuate uterus.

9- Allergy to copper (Wilson's diseases).

Insertion of IUCD:

A) Time of insertion:

1- Postmenstrual: → *advantages:*

a- Exclude pregnancy.

b- Insertion is easier as cervix is more patulous.

c- Post insertion spotting does not cause anxiety.

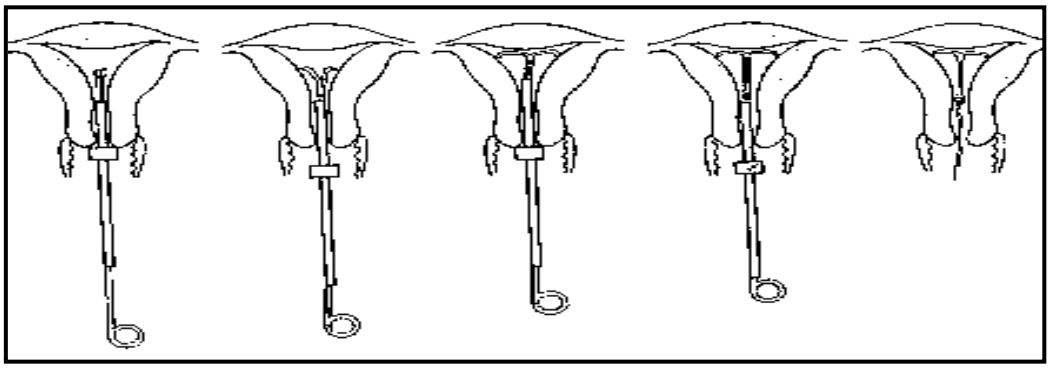
d- Post insertion discomfort is less.

2- Postpartum: (6-8 weeks) after delivery.

3- Post abortive (1-2 weeks) after abortion.

4- Post coital contraception.

B) Technique of insertion:



1- Examination of female genital system.
2- Sterilization of vagina and cervix after application of cusco speculum.
3- Holding anterior lip of cervix by volsellum.
4- Sounding of uterine cavity.
5- Unloading of loop inside uterus by either:
   - Pushing Method (Pushing)
   - Withdrawal method

C) Post insertion counseling:

1- Post insertion discomfort may occur as;
   a- Lower abdomen colic.
   b- Post insertion spotting.
   c- Increase mucoid discharge.
2- Patient should feel and not pull the nylon threads by her fingers in vagina especially after menses.
3- The woman should attend for check-up examination after one month and then annually.
4- The woman should ask for medical advice of :

P	A	I	N	S
↓	↓	↓	↓	↓
Period late	Abnormal pain	Infection	Not feeling well	String missing

Side effects, complication of IUCD:

- Bleeding
- Pain
- Pregnancy
- Pelvic infection
- Missed loop
- Male discomfort

I) <u>Bleeding</u>:

- Types of bleeding → Cyclic: menorrhagia
→ Acyclic: metrorrhagia

- Causes: → * ***Mechanical:*** - Large loop
- Wrong position

* ***Chemical:*** - ↑ vascularity.
- ↑ fibrinolysis

- Treatment: 1- Antifibrinolytic agents: e.g. tranexamic acid (cyclokapron)
2- Anti PG.
3- Ethamyselate (Dicynone)
4- If medical ttt failed → removal of loop.

II) <u>Pain</u>:

- Types : - Uterine cramps - Dysmenorrhea - Dyspareunia

- Causes: 1- Disparity between size of uterus and IUCD
2- Pelvic infection.
3- Ectopic pregnancy.
4- Perforation of uterus.

- Treatment : 1- Analgesics**:** e.g. Anti PG
2- Treatment of the causes

III) <u>Pregnancy</u>:

1- Intra-uterine pregnancy:

* <u>Complicated</u> by: - Abortion
- Septic abortion
- Preterm labour.
- Congenital anomalies (may be).

* <u>Treatment</u>**:** If threads of loop arevisible: gentle removal,
if not: removed with labour.

2- Extra-uterine ectopic pregnancy : ttt of ectopic pregnancy

IV- <u>Pelvic infection</u>:

- Types: 1- vaginitis. 2- cervicitis. 3- PID.

- ttt: Removal of loop with antibiotic therapy

V- Missed loop: (Inability to feel threads in vagina)

- Causes: 1- Insufficient instructions to female how to feel the threads.

2- Deep vagina or short fingers.

3- Expulsion.

4- Pregnancy.

5- Displacement of loop inside uterus.

6- Perforation of uterus.

- Management:

1- History: - Exclude pregnancy by no delayed cycles.

- History of abdominal pain and spotting during insertion of loop → suggest perforation.

2- P.V and bimanual examination: to exclude:

- Inability to feel threads due to deep vagina or short fingers.
- Pregnancy.

3- Sounding of uterus: if loop inside uterus, give positive click sound.

4- Plain X-ray on pelvis : - A-P view

- Lateral view with sound inside uterus

5- U/S

6- MRI

7- Combined hysteroscopy and laparoscopy for diagnosis and removal of loop.

8- Sometimes we do mini-laparotomy for diagnosis and removal of extra-uterine pelvic IUCD.

V- Hormonal Contraception

Steroid — GnRh-analogue

Steroid Contraception

Chemistry : The steroids used in hormonal contraception are:

- Synthetic estrogens: either ethinyl estradiol or mestranol.
- Synthetic progesterone (Progestagen or progestin) either 19-nortestosterone group or 17-∝- OH progesterone group.

Types of formulation:

A) Combined OCP: Pack = 21 tab each tab contains both E + P.

*** Monophasic:**

1- High dose estrogen compounds when (E) component of pills is more than 50µ / tab. e.g. Anovlar

2-Low dose estrogen compounds: when (E) component of pills is less than 50µ/tab e.g. Gynera, Microvlar.

***Biphasic:** 1st 10 tab of pack contains E+P in different doses than other 11 tab of pack.

*** Triphasic:** 1st 6 tab contain E + P in different doses other than follow 5 tab than follow 10 tab

*** Combined fixed dose pills (Monophasic):**

Class	E-dose	Trade name	E	P
High dose	80-100ug			
Moderate dose	50 ug	Anovlar primovlar	EE EE	Norethisterone 1 mg Norgestrel 0.5 mg
Low dose	30-35ug	Norminest Microvlar Nordette	EE EE EE	Norethisterione 0.5mg Levonorgestrel 150 ug Levonorgestrel 150 ug
Lowest dose	20 ug	Loestrin 20	EE	Norethisterone l mg

*** New progestagens in combined pills:**

1- Gestodene → Gynera = Gestodene 75 ug + EE 30 ug

2- Desogestrel → Marvilon = Desogestrel 150 ug + EE 30 ug.

3- Norgestimate → Cilest = Norgestimate 250 ug. + EE : 35 ug.

B) Sequential pills:

1st 14 days contain (E) only, while second 7 days contain (E + P) pills, this type of pill is removed from market since 1976 due to its major side effects of (E) alone pills → precancer effect.

C) Progestin only regimens:

Progestin only pills (Mini-Pills) | long acting injectable progestin (The shot) | Subdermal implants

1- Progestin only pills:

- Pills contain progestin only.
- Used continuously.
- Example: Microlut.

2- Long acting injectable progestin:

- Injections contain progestin only:
- *Examples:*

a- Depot medroxy progesterone acetate (DMPA) 150 mg injected I.M./3 months (Depo-provera).

b- Northisterone enanthate (Net-EN) 200 mg injected I.M/2 months (Noristerat).

3- Subdermal implants:

- Capsules containing long acting progestin are put through skin of medial aspect of the upper left arm.

*** Non- biodegradable:**

1- Norplant: 6 silastic capsules with levonergestrel

2- Norplant -2: 2 silastic rods with levonergestrel.

3-Implanon: 1 ethylene vinyl acetate (EVA) rod with 3-ketodesogestrel.

4- ST-1435 : 1 capsule

5- Uniplant: 1 silastic rod with nomegestrel acetate.

*** Biodegradable:**

1- Capronor: 1 capsule with levonorgestrel

2- Annuelle: 4 pellets with 90% norethisterone & 10% cholesterol.

3- BIDS- Net: Norethisterone.

4- BIDS- NOG: Norgestrel .

5- Biodegradable microsphere Containing norethisterome

Comparison between subdermal implants

Type	N	Container	(P)	Release/d	Duration
Norplant	6	Silastic	(36x6) mg levonorg-estrel	80 ug/d for few days ↓ 40 ug/d for 6-9 months ↓ 25 ugld	5Y
- Norplant-2	2	Silastic	(70x2) mg levonerg.	80 ugld.	3Y
- Implanion	1	EVA	67 mg 3-keto-desogestrel	40 ugld	3Y
- ST-1435	1	Silastic	40mg of P (ST-1435)	-	6-12months
- Uniplant	1	Silastic.	Nomegestrel acetate	-	1Y
- Capronor	1	Caprolactone Capsule	16 mg levonorgestrel	40 ugld	1Y
- Pellets	4	Pellets	90% norethisterone 10% cholesterol	-	1Y
- BIDS-Net		-	Norethisterone	-	-
- BIDS-NOG		-	Norgestrel	-	-
-Micro spheres		Microsphere	Norethisterone	-	-

- 2 types :

* Non biodegeradable capsules

- Norplant. - Norplant 2. - Implanon.

* Biodegradrable capsules:

- Microspheres. - Pellets.

Mechanism of action of steroid contraception:

A) E: The main site is central action

1- Inhibition of H-P-O axis → suppress ovulation.

2- It has effect on endometrium which become unsuitable for implantation.

3- ↑ Tubal motility.

B) P: The main site of action is peripheral:

1- It changes the architecture of cervical mucus from parallel lines into network pattern, so decrease sperm penetration, it is main action in progestin pills.
2- Antiestrogenic effect on uterus which become unsuitable for implantation.
3- ↓ tubal motility.
4- Suppress ovulation through central action/especially in large does of progesterone (Depoprevera).

Failure rate:

1- In COCP: 1/HWY
2- In progestin only pills: 2-4/HWY
3- In injectable progestins: 0.3/HWY.
4- In subdermal implants: <1 / HWY.

Indications of steroid hormonal contraception:

A)<u>Combined oral contraceptive pills</u>:

I- Contraceptive indications:

1-Newly married female.
2-Reliable method of contraception for most of females.

II- Non- contraceptive indications:

1- Dysfunctional uterine bleeding
2- Functional ovarian cysts.
3- Endometriosis
4- Premenstrual tension syndrome.
5- Spasmodic dysmenorrhea.
6- Regulation of cycle.
7- Postpone menstruation.
8- Family history of cancer ovary.

B- <u>Progestin only system</u>:

I- Contraceptive indications:

1- During lactation.

2- In old women.

3- If there is contraindications to estrogens.

4- In diabetic patient.

II- Non-contraceptive indications: for treatment of hyperestrogenic state.

Contraindications of steroid contraception:

I- <u>Absolute contraindications</u>:

1- History of thromboembolic diseases.

2- Known or suspected cancer breast.

3- Known or suspected estrogen dependant tumour.

4- Known or suspected pregnancy.

5- Abnormal liver function.

6- Undiagnosed abnormal uterine bleeding.

7- Obstructive jaundice.

8- Congenital hyperlipidaemia.

9- Old age.

10- Cerebrovascular stroke.

II- <u>Relative contra indications</u>:

1- Migraine, headache.

2- Hypertension.

3- D.M.

4- Varicose veins.

5- Surgery.

Side effects and complication:

1- Nausea and may be vomiting due to ↑ (E).

2- Headache and depression → ↑ (P).

3- Leucorrhea due to ↑ (E).

4- Cosmotic effect → ↑ (P) + (E).

weight gain acne hair loss

5- Menstrual changes:

- Spotting and break through bleeding
- Hypomenorrhea, and may be amenorrhea

6- Metabolic effect:

a- CHO: Hyperglycemia:

(E) : ↑ blood glucose due to↑ glycogenolysis

(P): ↑ insulin resistance

b- lipids: ↑ incidence of atherosclerosis

c- proteins: ↑ clotting factors → hypercoagulability

7- On exist pregnancy: especially in 1st trimester (Vactrel syndrome).

V: vertebral bodies congenital anomalies.

A: anorectal congenital anomalies.

C: cardiac congenital anomalies.

T: trachea congenital anomalies.

R: renal congenital anomalies.

E: esophagus congenital anomalies.

L: limb congenital anomalies.

8- On breast milk production: (E) → ↓ milk production.

N.B.1: post-pill Amenorrhea: amenorrhea lasts > 3 cycles after stopping of the pills due to either suppression of hypothalamic axis (treated by clomiphene) or increase of prolactin (treated by Bromocryptine).

N.B.2: side effects of progestin are mainly:

- Menstrual disturbances -Headache - Depression - Weight gain.

IV- Sterilization:

Male :

1- Vas ligation

2- Vasectomy

Female

1- Hysterectomy

2- Tubal sterilization

Methods of tubal sterilization

[1] Mini- laparotomy:

1- Madlener: crushing & ligation of a tubal loop by non-absorbable suture.

2- Irving: excision of part of tube, then: medial end embedded into myometrium & lateral end between 2 layers of broad ligament

3- Cook: as Irving but medial end in round ligament.

4- Ushida: as Irving but medial end in brood ligament.

5- Pomeroy: loop of tube is isolated, doubled ligated & then it is cut.

6- Knoener: Fimbriectomy

7- Parkland: Mesosalpinx is perforated in avascular area, tube ligated by chromic O & segment excised.

[2] Laparoscopic tubal ligation

1- Fulguration: by electrocauterization.

2- Clips: Hulka

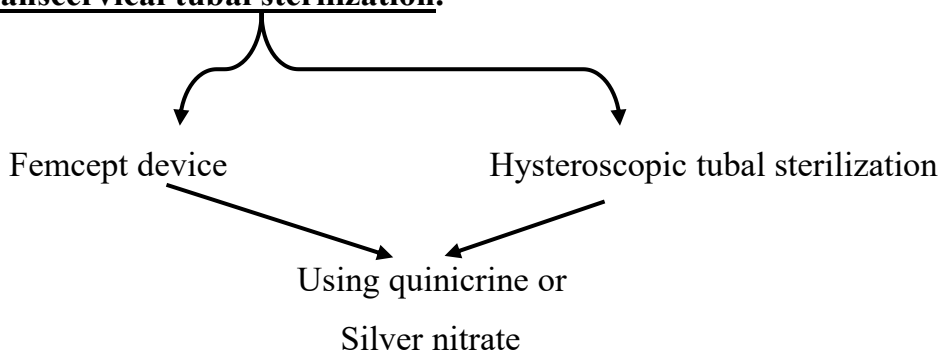
3- Rings: Falop ring

4- Laser beam.

[3] Culdoscopic tubal ligation

[4] Posterior colpotomy tubal ligation.

[5] Transcervical tubal sterilization.



These are permanent methods of contraception indicated in woman with ↑ age (> 40 years old) with a disease which affects woman's life if she becomes pregnant and no other methods of contraception can be used.

Measures of contraceptive effectiveness

1- Theotrical (Method) effectiveness:

Consider only pregnancy which occur when method is used regularity & correctly.

2- Classic use effectiveness:

Consider only pregnancy which occur when method is used with regularity & not necessary correctly.

3- Extended use effectiveness:

Consider pregnancy due to method failure + failure due to incorrect use + failure due to irregular use.

Formula used to measure any type of failure:

1- Pearl index: number of pregnancy /100 ♀ used this method for 1 year = Hwy. Does not take into account duration of using the method.

2- Life-table analysis: number of pregnancy /100 ♀ used the method for 1 year (1st year of use).

Method	Theotrical failure	Use failure
- Tubal ligation	0.0	0.1
- Vasectomy	0.0	0.1
- DMPA/ 3 Months	0.0	0.25
- COCP	0.1	1
- POP	1	2-4
- IUCD:		
Group I:	2-4	
Lippes loop		
Cu 7		
Cu T 200		
Group II:	1-2	
Cu T 220		
Nova T		
Group III:	< 1	
CuT380		
Mutiload 375		
levo nova		
- Male condom	1.5	5-15
- Female barrier	5	20
- Spermicidal	3	30
- Coitus Interruptus	5	20-30
- Safe period	2	20

N.B.: Methods of male contraception:

1- Coitus interruptus.
2- Male condom.
3- Hormonal contraception (Male pills). Drugs which ↓ FSH in male so → ↓ sperm count e.g. inhibin.
4- Male sterilization: Vasectomy.

N.B.: Post coital contraception:

Definition: Prevention of implantation of fertilized ovum after mid cycle unprotected coitus.

Methods:

1- High dose estrogen.
2- Mini pills (progesterone).
3- Combined pills : e.g. Anovlar 2 tab repeated after 12 hr in 1st 72 hr after unprotected coitus.
4- Danazol in high dose.
5- Antiprogesterone (RU 486) = Mifepristone
6- IUCD after coitus.
7- Menstrual extraction by suction extractor.
8- Post coital vaginal douching by any antiseptic solution or acetic acid.

N.B: Contraception to special groups:

1- Newly married couples: They can use either
- Safe period (best method) - Male condom
- Oral contraceptive pills - POP.

2- With lactation: - I.U.C.D. - Progestin only pills (POP).

3- Old age (above 35 years old).
- I.U.C.D. - Depoprovera.
- Sterilization (controversy).

4- Women with cardiac disease.
- Progestin only system – Male condom.
- I.U.C.D or combined pills are not preferred.

5- Women with D.M. : as cardiac patient.

* **Q- For Written Paper:**

1- Contraception for newly married couples.

2- Contraception for ♀Above forties.

3- Contraception for cardiac pt.

4- Contraception for diabetic pt.

5- Recent trends in spermicidal.

6- Complications of pregnancy with loop in situ.

7- Management of pelvic pain with IUCD.

8- Missed loop.

9- Dose & preparations of hormonal contraception's.

10- Progestin only systems.

11- Subdermal implants.

12- Injectable contraceptions.

13- Male contraceptions.

14- Post coital contraceptions.

* **Oral Q:**

1- Management of missed pills:

* لو نسيت حبة تاخذها بمجرد أن تتذكر وتأخذ الحبة الأخرى فى الموعد المعتاد
* لو نسيت 2 حبة تأخذ 2 حبة فى اليومين القادمين مع اخذ احتياط أخر لمدة 7 أيام0
* لو نسيت أكثر من 2 حبة :
  - 2 حبة لمدة تعوض المنسي مع اخذ احتياط0
  - أو يترك دم The Cycle وتبدأ شريط أخر من خامس يوم الدورة0

2- Drugs ↓ effects of COCP leads to: - Failure or break through bleeding.

* All enzyme inducer drugs on liver like anticonvalsants, rifampian, chlorpromazine, spironolactone.
* Ampicillin: as change GIT flora.

3- Drugs ↑ effects of COCP: Vit. C, cotrimoxazole.

4- B.T.B: irregular bleeding in 1st few months after start of pills may be due to

- ↑ progestin or ↓ EE

Treatment: (1) ↑ EE (2) Change progestin to another preparation .

5- Inject able contraceptions:

- **P-only :** DMPA , Norigest
- **E+P :** Mesigyna (50 mg norethisterore acetate + 5mg E. valarate) حقن شهر
- **Others:** GnRh-A حقن / شهر

6- Measures of contraceptive effectiveness.

7- Post coital contraceptions.

8- Types of female sterilizations.

9- Post tubal ligation syndrome: months after tubal ligature → congestion: menorrhagia & metrorhagia.

10- Side effects of progestin only pills.

11- Mechanism of action of COCP.

12- Side effects of IUCD & how to managed.

13- Mechanism of action of cu T-loop.

14- How to ↑ efficacy of male condom.

15- Non- contraceptive benefits of male condom.

16- Measure of safe period.

N.B. <u>Recent IUD</u>:

1- Gynefix: frameless IUD, try to overcome expulsion & heavy menses. Consists of 6 copper cylinder 5 mm X 2.2 m each on a monofilament polypropylene thread. The upper & lower cylinders are crimped on to thread to secure placement. At the upper end of thread is a knot which is implanted into the fundal myometium acting as anchor.

2- Grnefix post portum (PP): Thicker polypropylene thread and cone shaped 4 mm x 5mm biodegeradable body immediately below the knot. This core gradually biodegerade over 2-5 w & therefore moulds to the involuting uterus.Gynefix PT has the same thread as PP with slightly bigger anchoring knot compared to the standard model to enhance the retention force.

[7] Hormones in Gynecology:

[A] Indications of use of E:

I. Improve vulvovaginal atrophic cases:

1- Prepubertal senile vulvovaginitis.
2- Atrophic chronic vulvar epithelium dystrophies.
3- Trophic ulcers of prolapse.
4- Conservative treatment of prolapse, S.I. and fistula.
5- Postmenopausal pruritus vulvae.

II- To replace E↓ (ERT):

1- Ovarian failure either aplasia or hypoplasla
2- Menopausal syndrome after bilateral salpingoophrectomy
3- In some cases of amenorrhea e.g. hypoplastic uterus.

III. Some hormonal disturbances:

1- Spasmodic dysmenorrhea:
2- Dysfunction uterine bleeding e.g. threshold bleeding.
3- Cx. factor of infertility or with clomid therapy.

IV. For contraception

[B] Indications of use of progesterone:

1. In all hyperestrogenic state e.g. dysfunction uterine bleeding endometriosis, fibroid.
2. Infertility: luteal phase defect.
3. For contraception.

[C] Gonadotropin Releasing Hormone analogue: (GnRH- a):

Replacement of amino acid in position 6 and/ or 10 of the native GnRh-molecule (decapeptide) producing.

Synthetic superactive analogue that resists degradation by enzymes and hence a prolonged half life.

- Indications**:**

1-Endometriosis
2--Dysfunction uterine bleeding.

3- Fibroids

4-Precocious puberty.

5- Premenstrual tension syndrome.

6-Hirsutism .

7- Ovarian and breast cancer.

8- In case of P.C.O. : for induction of ovulation,

- Methods administration:

1- Intranasal spray
2- Subcutaneous injection (in anterior abdominal wall)
3- I. M.I.

- Side effects:

1- Symptoms of hypoestrogenism e.g. hot flushes, vaginal dryness, osteoporosis, insomnia.
2- Anaphylactic reaction.

Local irritation at site of administration.